Efficient GAN-Based Anomaly Detection

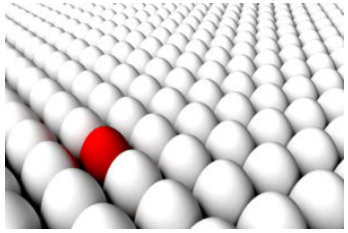
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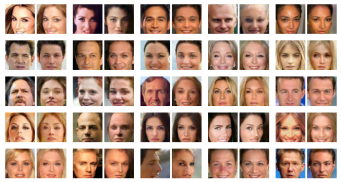
April 9, 2018

Context

Problem: Unsupervised anomaly detection

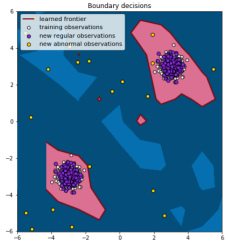


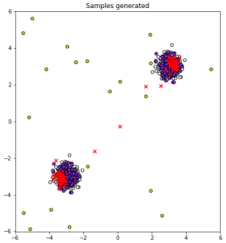
We wish to effectively model the data generating distribution ...





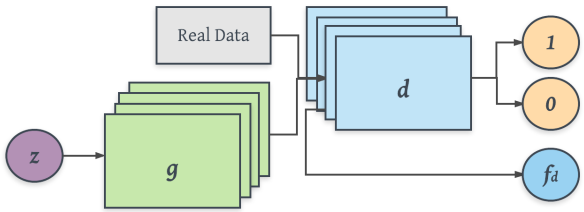
... and derive a statistically sound decision criteria for anomaly detection





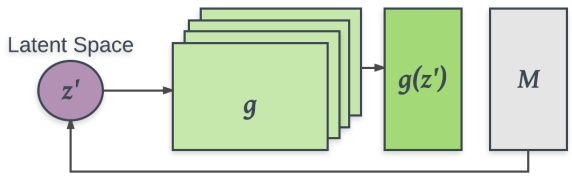
Context

Generative adversarial networks (GANs) [5] can model complex and high-dimensional distributions, particularly over natural images [9]





A GAN that has been well-trained to fit the distribution of normal samples should be able to reconstruct a normal sample from a certain latent representation and also discriminate the sample as coming from the true data distribution.



Overview

Related Work - Anomaly Detection

Related Work - Deep Learning for Anomaly Detection

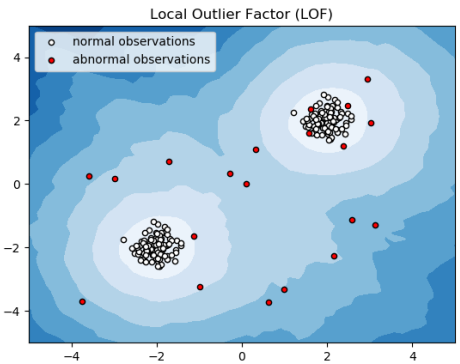
Related Work - AnoGAN

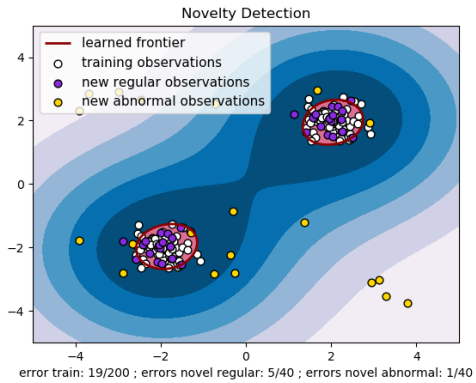
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Improvements of the model

Anomaly Detection

- Clustering approaches or nearest neighbor methods [14, 18], such as LOF method, and one class classification approaches that learn a discriminative boundary around normal data, such as one-class SVMs [15].





- Another class of methods uses fidelity of reconstruction to determine whether an example is anomalous, and includes Principal Component Analysis (PCA) and its kernel and robust variants [7, 10, 3].

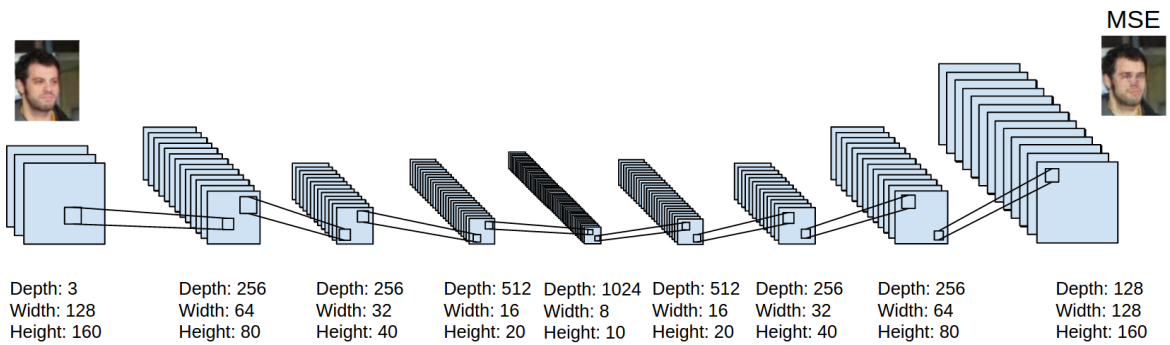
Shortcomings of such Anomaly Detection methods

Curse of dimensionality with such feature construction methods:

- Concentration of scores and distances: derived values such as distances become numerically similar
- Irrelevant attributes: in high dimensional data, a significant number of attributes may be irrelevant
- Exponential search space: the search space can no longer be systematically scanned

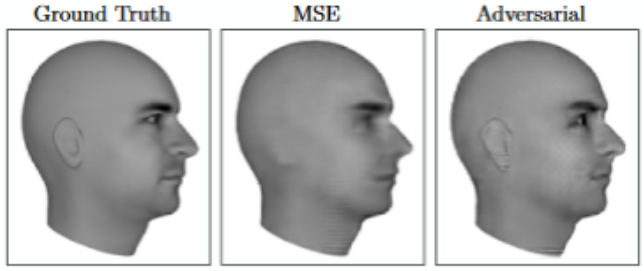
Deep Learning for Anomaly Detection

More recent works use deep neural networks, which do not require explicit feature construction unlike the previously mentioned methods.
Autoencoders, variational autoencoders [1, 17], energy based models [16] and deep autoencoding Gaussian mixture models [2] have been explored for anomaly detection.



Shortcomings of Auto-encoders based Anomaly Detection

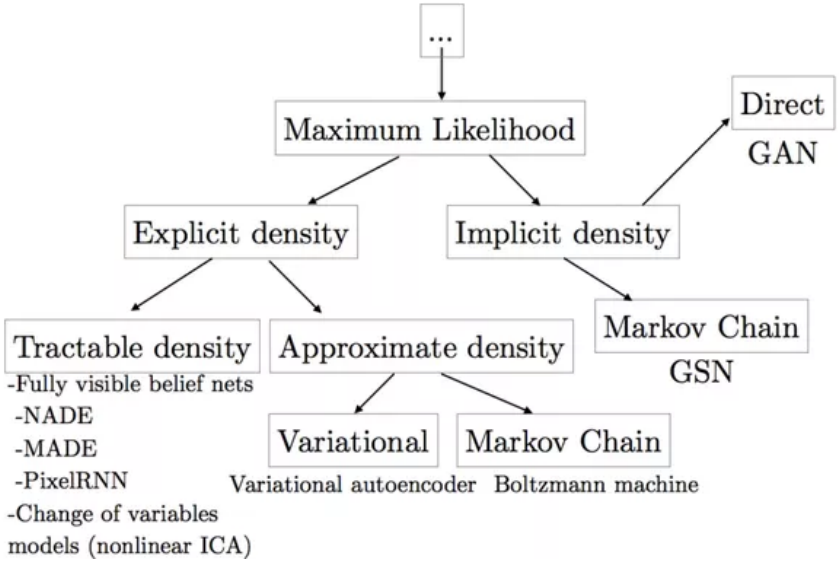
- L-k losses: Auto-encoders tend to have poor reconstructions on high dimensional data



- Priors in the latent space: the log-likelihood lower-bound in the loss function implies to know the density function and to be efficiently computable.
  (Ex: $\mathbb{E}_{q(z|x)}[\log p(x|z)] - KL(q(z|x)||p(z))$) The approximate posterior tends to be simple and tends to not "fill in" the space covered by the prior.
- Adversarial learning frees the learning from explicitly defining a density function and has empirically more expressive power.

Background - GANs

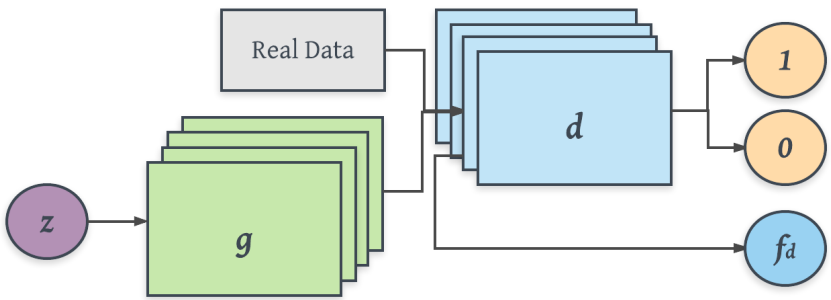
Deep generative models that can learn via the principle of maximum likelihood differ with respect to how they represent or approximate the likelihood [6]



GANs are an implicit model that sample directly from the distribution represented by the model.

Background - GANs

Generative Adversarial Networks: Two players adversarial game where a generator samples data from a noise seed in a latent space, and discriminator tries to distinguish real data from fake generated data.

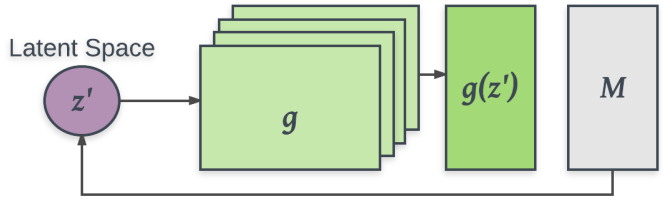


$\min_G \max_D V(D, G)$, with $V(D, G)$ defined as

$$V(D, G) = \mathbb{E}_{x \sim p_X} [\log D(x)] + \mathbb{E}_{z \sim p_Z} [\log (1 - D(G(z)))] .$$

Related Work - AnoGAN

- Exploiting the arrangement of the latent space by projecting images into it, and using this as a representation for discriminative tasks and outlier detections.



- Stochastic gradient descent (Adam optimizer) is used to find z' such that $z' = \arg\, min_z(L_{tot}(z))$, to "invert" an input data x

$$L_{total}(z') = \alpha L_G(z') + (1 - \alpha)L_D(z')$$

- $L_G(z') = \|x - G(z')\|_k$, k is an integer (1 or 2)
- $L_D(z') = \sigma(D(G(z')), 1)$ is the output of the discriminator
- or $L_D(z') = \|f_D(x) - f_D(g(z'))\|_k$, where f_d is an intermediate layer in the discriminator, feature matching trick [11]

Related Work - AnoGAN

The anomaly score is derived from the mapping loss function:

$$A(x) = \alpha \|x - x'\|_k + (1 - \alpha) L_D(x, x')$$

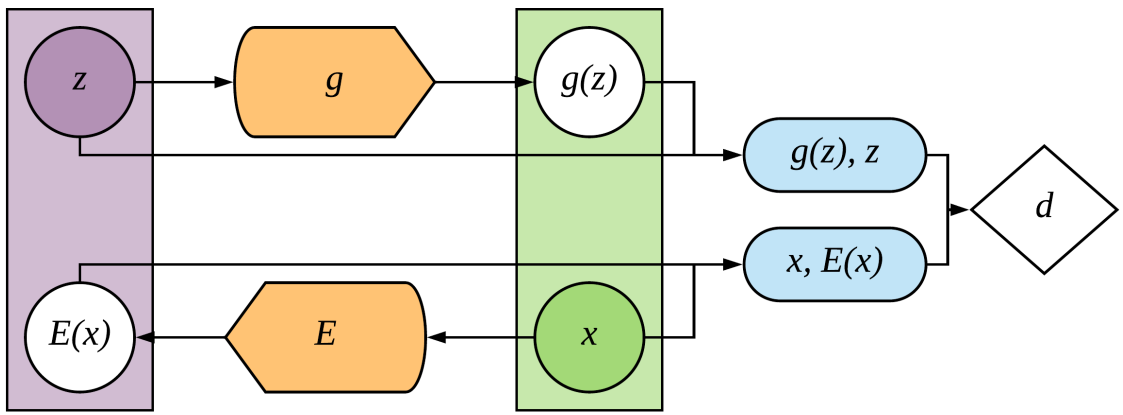
where x is the input data, x' is the reconstructed image
L_d feature matching or output of the discriminator
α is a weight parameter, in experiments $\alpha = 0.9$

Shortcomings:

- Inaccurate latent recovery of latent representation in a non convex space: initialize it several times or use stochastic clipping [8]
- Even with precise latent recovery of latent vectors, the computational cost of inverting the generator makes it virtually unrealistic for practical applications in research and industry...

Enabling inference with GANs

Instead of inverting the generator, we learn the inverse mapping (by also projecting data back into the latent space): the model jointly learns a generation network and an inference network using an adversarial process. [13, 4]



The generator g maps samples from latent variables to the data space while the encoder E maps training examples in data space to the space of latent variables.

$\min_{G,E} \max_D V(D, G, E)$, with $V(D, G, E)$ defined as

$$V(D, G, E) = \mathbb{E}_{x \sim p_X} [\log D(x, E(x))] + \mathbb{E}_{z \sim p_Z} [\log (1 - D(G(z), z))]$$

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Having trained a model to yield G, D and E, we then define a score function $A(x)$ (as in [12]) that measures how anomalous an example x is, based on the convex combination of a reconstruction loss L_G and a discriminator-based loss L_D:

$$A(x) = \alpha L_G(x) + (1 - \alpha) L_D(x)$$

where $L_G(x) = ||x - G(E(x))||_k$ and $L_D(x)$ is also defined in two ways. First, using the output of the discriminator $L_D(x) = \sigma(D(x, E(x)), 1)$, or $L_D(x) = ||f_D(x, E(x)) - f_D(G(E(x)), E(x))||_1$, with f_D returning the layer preceding the logits for the given inputs in the discriminator.

Experiments - MNIST

- We generated 10 different datasets from MNIST by successively making each digit class an anomaly and treating the remaining 9 digits as normal examples.
- The training set consists of 80% of the normal data and the test set consists of the remaining 20% of normal data and all of the anomalous data. All models were trained only with normal data and tested with both normal and anomalous data.
- Our model significantly outperforms the auto-encoder baseline. Likewise, our model outperforms AnoGAN (Figure 1), but with approximately 800x faster inference time (Table 2).

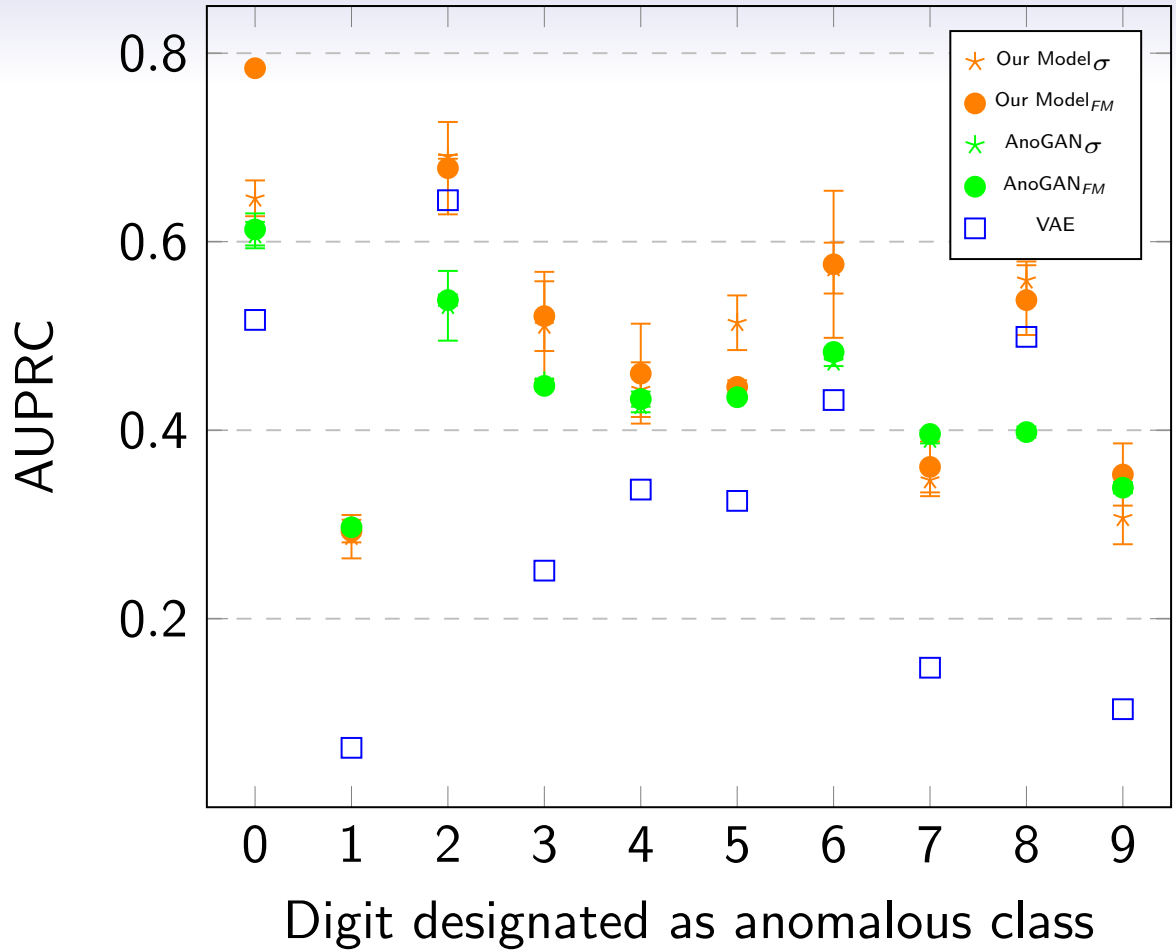


Figure: Performance on MNIST measured by the area under the precision-recall curve. VAE data were obtained from [1]. Error bars show variation across 3 random seeds. FM and σ respectively denote feature-matching and cross-entropy variants of L_D used in the anomaly score.

Experiments - KDD99

- We evaluated our method on this network activity dataset to show that GANs can also perform well on high-dimensional, non-image data. We follow the experimental setup of [16, 2]
- For training, we randomly sampled 50% of the whole dataset and the remaining 50% of the dataset was used for testing. Only data samples from the normal class were used for training.
- The 20% of samples with the highest anomaly scores $A(x)$ are classified as anomalies (positive class).
- Our method is overall highly competitive with other state-of-the art methods and outperforms AnoGAN with a 700x to 900x faster inference time (Table 2)

Experiments - KDD99

Table: Performance on the KDD99 dataset. Values for OC-SVM, DSEBM, DAGMM values were obtained from [16, 2]. Values for AnoGAN and our model are derived from 10 runs.

Model	Precision	Recall	F1
OC-SVM	0.7457	0.8523	0.7954
DSEBM-r	0.8521	0.6472	0.7328
DSEBM-e	0.8619	0.6446	0.7399
DAGMM-NVI	0.9290	0.9447	0.9368
DAGMM	**0.9297**	0.9442	0.9369
AnoGAN$_{FM}$	0.8786 $\pm$ 0.0340	0.8297 $\pm$ 0.0345	0.8865 $\pm$ 0.0343
AnoGAN$_{\sigma}$	0.7790 $\pm$ 0.1247	0.7914 $\pm$ 0.1194	0.7852 $\pm$ 0.1181
Our Model$_{FM}$	0.8698 $\pm$ 0.1133	0.9523 $\pm$ 0.0224	0.9058 $\pm$ 0.0688
Our Model$_{\sigma}$	0.9200 $\pm$ 0.0740	**0.9582 $\pm$ 0.0104**	**0.9372 $\pm$ 0.0440**

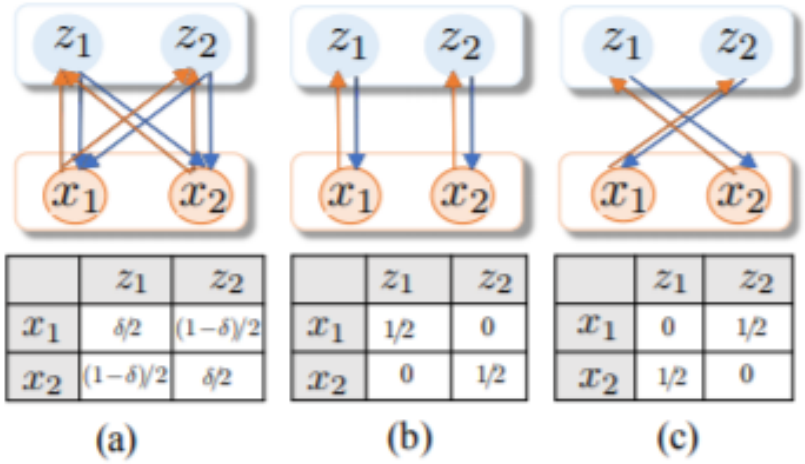
Experiments - Inference

Table: Average inference time over 100 batches (ms)

Dataset	L_D	AnoGAN	Our Model	Speed Up
MNIST	σ	6657	8	$\sim$830
	FM	7260	9.4	$\sim$ 770
KDD	σ	2578	2.7	$\sim$950
	FM	3527	5,3	$\sim$660

Shortcomings of this model

Non-identifiability issues:



	z_1	z_2
x_1	$\delta/2$	$(1-\delta)/2$
x_2	$(1-\delta)/2$	$\delta/2$

(a)

	z_1	z_2
x_1	1/2	0
x_2	0	1/2

(b)

	z_1	z_2
x_1	0	1/2
x_2	1/2	0

(c)

It is possible to obtain an undesirable solution to the minmax objective. For instance, i) one with poor reconstruction ability or ii) one where a single instance of z can potentially map to any possible value in x , e.g., in Figure 1(a) with $\delta = 0.5$, z_1 can generate either x_1 or x_2 with equal probability.

Shortcomings of this model

Poor reconstructions:



Figure: Reconstructions over SVHN

Improvement of the method

We define the joint distribution $\pi(x, z)$ where x is in the data space and z in the latent space. Let's consider the conditional entropy:

$$H^{\pi}(x|z) = -\mathbb{E}_{\pi(x,z)}\left[\log \pi(x|z)\right]$$

$$H^{\pi}(z|x) = -\mathbb{E}_{\pi(x,z)}\left[\log \pi(z|x)\right]$$

We define a new objective function:

$$V'(D, G, E) = V(D, G, E) + L_{CE}(G, E)$$

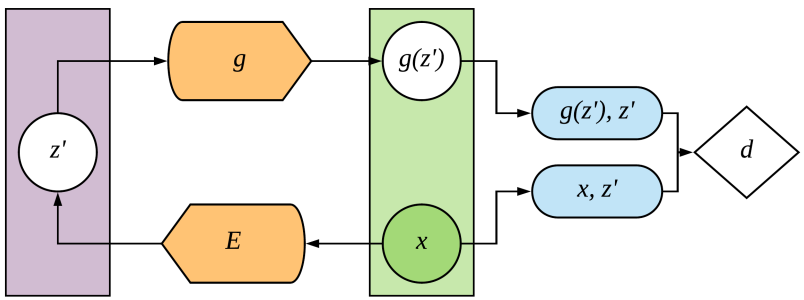
,
where $L_{CE}(G, E) = H^{\pi}(z|x) + H^{\pi}(x|z)$

Improvement of the method

We use both non-adversarial and adversarial approaches to learn desirable matched joint distributions for unsupervised tasks.

Non Adversarial:

$$L_{CE} = ||x - G(E(x))||_2 + ||E(x) - E(G(E(x)))||_2$$



Adversarial:

$$L_{CE} = \mathbb{E}_{x\sim p_X}\left[\log D_{xx}(x,x)\right] + \mathbb{E}_{x\sim p_G(\cdot|z),z\sim p_E(\cdot|x)}\left[1 - \log D_{xx}(x,x')\right]$$
$$+ \mathbb{E}_{z\sim p_z}\left[\log D_{zz}(z,z)\right] + \mathbb{E}_{z\sim p_E(\cdot|x),x\sim p_G(\cdot|z)}\left[1 - \log D_{zz}(z,z')\right]$$

Where D_{xx} and D_{zz} are additional discriminator networks.

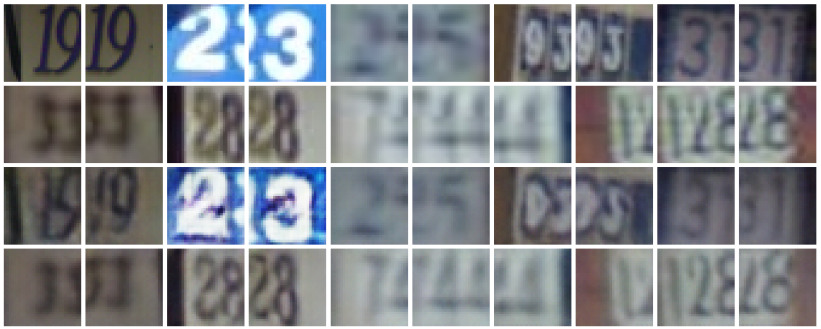
First experiments

Improved reconstructions

Non adversarial:



Adversarial:



Efficient GAN-Based Anomaly Detection

Thank you!


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